

Exam ratio practical question

Que.1) The following trading or profit and loss account of fantasy ltd for the year 31-3-2017 is given bellow.

Particular	Amount	Particular	Amount
To opening stock	76250	By sales	500000
To N purchase	315250	By closing stock	98500
To carriage and fright	2000		
To wages	5000		
To gross Profit c/d	200000		
	<u>598500</u>		<u>598500</u>
To administrative expenses	101000	by gross Profit c/d	200000
To selling and dist. Expenses	12000	By non operating incomes: -	
To non operating expenses	2000	Dividend on share	3750
To Financial expenses	7000	Interest on securities AF	1500
		Profit on sale of share	750
To net profit c/d	84000		
	<u>206000</u>		<u>206000</u>

1) Gross profit ratio = gross profit/ net sales *100

Gross profit ratio = $200000/500000*100$

40%

2) Net profit ratio = Net profit/ Net sales*100

Net profit ratio = $84000/500000*100$

16.8%

3) Operating ratio = cost of goods sold+ operating expenses/ net sales*100

A) Cost of goods sold = opening stock + purchases + wages + carriage – closing stock

Cost of goods sold = $76250+315250+2000+5000-98500$

Cost of goods sold = 300000

B) Operating expenses= office + administrative expenses

$$\text{operating expenses} = 101000 + 12000 \\ = 113000$$

$$\text{Operating ratio} = 300000 + 113000 / 500000 * 100 \\ 82.6\%$$

Que. 2) calculate liquidity ratio

<u>Current liabilities</u>	<u>50000</u>
<u>Current assets</u>	<u>80000</u>
<u>Inventories</u>	<u>20000</u>
<u>Advance Tax</u>	<u>5000</u>
<u>Prepaid expenses</u>	<u>5000</u>

- **Current ratio = current assets / current liabilities**

$$\text{Current ratio} = 80000 / 50000$$

Ans. 1.6:1

- **liquid ratio or Quick ratio = liquid assets/ current liabilities**

Liquid assets= current assets – inventories – advance tax – prepaid expenses

$$\text{liquid assets} = 80000 - 20000 - 5000 - 5000 \\ 50000$$

$$\text{Liquid ratio or Quick} = 50000 / 50000$$

Ans. 1:1

Que.3) X Ltd has a current ratio of 3.5:1 and quick ratio 2:1 .if excess of current assets over quick assets represented by inventories is Rs. 24000. Calculate current assets and current liabilities.

Solution: -

inventory = current ratio – quick ratio

$$? = 3.5 - 2$$

$$\text{inventories} = 1.5$$

If inventories is 1.5, than current assets = 3.5

If inventories is 1, than current assets = 3.5/1.5

If inventories is 24000, than current assets = $3.5/1.5 \times 24000 = 56000$

As current ratio is 3.5:1 and current assets is Rs. 56000

Current liabilities = $56000/3.5$

Ans. 16000

Que. 4) The balance sheet of XYZ Company is given below:

Liabilities	Amount	Assets	Amount
Equity share capital	250	Fixed assets	400
General reserve	280	Investment	50
P&L A/C (current year)	30	Stock	460
Secured loans ----(long term)	300	Debtors	460
Secured loans---- (short term)	360	Cash in hand	10
Creditors	150	Misc expenditure – (not written off	20
Other liabilities	50		
	1400		1400

Additional information

- From the profit and loss account Rs.90 lacs was transferred to General Reserve during the year
- Interest cost amounted to Rs. 120 Lacs
- Taxation @ 40%

You are required to calculate 1) Debt Equity Ratio 2) Current Ratio 3) interest coverage ratio

1) Debt Equity Ratio

Debt equity ratio = long term debts /equity (shareholder funds)

Long term debts = Secured loans (long term)

Equity (shareholder funds) = Equity share capital + General reserve + P&L A/C (current year)

Equity (shareholder funds) = 250+280+30
=560

Debt equity ratio = 300/560

0.53:1

2) Current Ratio

Current ratio = current assets / current liabilities

Current assets = investment + stock + debtors + cash in hand

Current assets = 50+460+460+10/ 360+150+30

Current assets = 980/540

= 1.8:1

3) Interest coverage ratio

Interest coverage ratio = net profit before interest and income tax/ fixed income charges

Current year Profit = 30 Lacs

+ Transfer to general reserve 90 Lacs

Profit after tax 120 Lacs

Tax is 40%

Profit after tax is 60 the profit before tax = 100

1-----100/60

120-----100/60*120
=200 Lacs

Profit before interest and tax = 200+120 = 320

Interest coverage ratio = net profit before interest and income tax/ fixed income charges

320/120 = 2.66 times

Que .5) The following balance sheet of a company as on 31st March 2018

Liabilities	Amount	Assets	Amount
Share capital	200000	Land & building	140000
Profit & loss A/C	30000	Plant & machinery	350000
General reserve	40000	Stock	200000
12% debentures	420000	Sundry debtor	100000
Sundry creditors	100000	Bills receivables	10000

Bills payable	50000	Cash in hand	40000
	<u>840000</u>		<u>840000</u>

Calculate

a) Current ratio b) quick ratio c) inventory to working capital

a) Current ratio = current assets / current liabilities

Current assets = Stock + Sundry debtor + Bills receivables + Cash in hand

Current assets = 200000+ 100000+ 10000+ 40000
= 350000

Current liabilities = Sundry creditors+ Bills payable

Current liabilities = 100000+50000
=150000

**Current ratio = 350000/150000
=2.33:1**

b) Quick ratio = liquid assets/ current liabilities

Liquid assets = current assets – stock

Liquid assets = 350000-200000

Liquid assets = 150000

**Quick ratio = 150000/ 150000
=1:1**

c) Inventory to working capital = average inventories / net working capital *100

average inventories = 200000

Net working capital = current assets – current liabilities

Net working capital = 350000-150000 = 200000

Inventory to working capital = average inventories / net working capital
*100

Inventory to working capital = 200000/200000*100

100%

Que. 6) The following balance sheet of Punjab Ltd. As on 31st march
2002

Liabilities	Amount	Assets	Amount
Equity share capital	40000	Plant & machinery	24000
Capital reserve	8000	Land & building	40000
Loan 8%	32000	Furniture	16000
Creditors	16000	Stock	12000
Bank overdraft	4000	Debtors	12000
Taxation of current year	4000	Investment short term	4000
Taxation of future	4000	Cash in hand	12000
Profit & loss A/c	12000		
	<u>120000</u>		<u>120000</u>

Calculate 1) current ratio 2) quick ratio c) debt equity ratio

d) Proprietary ratio

1) Current ratio = current assets / current liabilities

Current assets = Stock+ Debtors+ Investment short term+ Cash in hand

Current assets = 12000+12000+4000+12000
= 40000

Current liabilities = Creditors+ Bank overdraft

Current liabilities = 16000+4000
=20000

$$\text{Current ratio} = 40000 / 20000$$

$$2:1$$

2) quick ratio = liquid assets / current liabilities

$$\text{Liquid assets} = \text{current assets} - \text{stock}$$

$$= 40000 - 12000$$

$$= 28000$$

$$\text{Quick ratio} = 28000 / 20000$$

$$1.4:1$$

3) debt equity ratio = debt / equity

Or

Long term debt / share holder funds

$$\text{Long term debt} = \text{Loan } 8\% + \text{taxation future}$$

$$= 32000 + 4000$$

$$= 36000$$

Share holder = Equity share capital + Profit & loss A/c + Capital reserve

$$= 40000 + 12000 + 8000$$

$$= 60000$$

$$36000 / 60000$$

$$0.6:1$$

Que.7) calculate liquidity ratio from the following

Current liabilities	50000
Current assets	80000
Inventories	20000
Advance Tax	5000
Prepared expanses	5000

Solution. Liquidity Ratio = Liquid Assets / Current Liabilities

$$\text{Liquidity Assets} = \text{Current assets} - (\text{Inventories} + \text{Prepaid expenses} + \text{Advance tax})$$

$$= \text{Rs. } 80,000 - (20,000 + 5,000 + 5,000)$$

= Rs. 50,000

Liquidity Ratio = 50,000/50,000

=1:1

Que.8) calculate current ratio from the following

Total assets	300000
Non- current liabilities	80000
Shareholder' funds	200000
Non- current assets	
Fixed assets	160000
Non- Current investment	100000

Solution.

Current Ratio =Current Assets/Current Liabilities

Total assets = Non-current assets + Current assets

Total assets = fixed assets + non-current investment + current assets

300000 = 160000+100000+ current assets

3,00,000 = 2,60,000 + Current assets

Current assets = 3,00,000- 2,60,000 = 40,000

Total assets = Equity and Liabilities

=Shareholder' Funds + Non-current liabilities + Current liabilities

3,00,000 = 2,00,000 - 80,000 + Current Liabilities

Current liabilities = 3,00,000 - 2,80,000

= 20,000

= 40,000/20,000=2:1

Que.9) A trader carries average inventories of Rs. 40000. His inventory turnover ratio is 8 times. If he sell goods at a profit of 20% on revenue from operation. Find out the gross profit.

Solution.

Inventory turnover = cost of goods sold / average inventory

8 = cost of goods sold /40000

$8 \times 40000 = \text{cost of goods sold}$

Cost of goods sold = 320000

profit of 25% on cost of goods sold (since 20% on sales)

gross profit = $3,20,000 \times 25\%$

= Rs.80,000

Que. 10) using information is given by a company from its books of accounts as on 31st march 2017

Particular	Amount
Inventories	100000
Total current assets	160000
Shareholder's funds	400000
13% debenture	300000
Current liabilities	100000
Net profit before tax	3561000
Cost of revenue from operation	500000

- **Calculate**
 1. **Current ratio**
 2. **Liquidity ratio**
 3. **Debt equity ratio**
 4. **Interest Coverage Ratio**
 5. **Inventory Turnover Ratio**

Solution:

1) Current Ratio= Current Assets /Current Liabilities

= $1,60,000 / 1,00,000$

= 1.6 : 1

2) Liquid Ratio= Liquid Assets/ Current Liabilities

Liquid Assets=Current assets– Inventory

= $1,60,000 - 1,00,000$

=Rs. 60,000

= $60,000 / 1,00,000$

= 0.6 : 1

- 3) **Debt-Equity Ratio**= Long-term Debts /Shareholders' Funds
 = 3,00,000 / 4,00,000
 = 0.75 : 1
- 4) **Net Profit before Interest**=Net Profit before Tax + Interest on Long term Debts
 = 3,51,000 + (13% of Rs. 3,00,000)
 =Rs. 3,51,000 + Rs. 39,000
 = Rs. 3,90,000
- 5) **Interest Coverage Ratio**= Net Profit before Interest & Tax /Interest on Long Term Debts
 =Rs. 3,90,000 / 39,000
 = 10 times
- 6) **Inventory Turnover Ratio** = Cost of Revenue from Operations/ Average Inventory
 =. 5,00,000 / 1,00,000
 = 5 times

Que. 11) from the data calculate

1. Gross profit ratio
2. Net profit ratio
3. Return on total assets

Sales	2520000
Cost of sales	1920000
Net profit	360000
Inventories	800000
Current liabilities	600000
Other current assets	760000
Fixed assets	144000
Net worth	1500000
Debt	900000

Solution:

1. Gross Profit Ratio = (GP/ Net Sales) * 100

Gross profit = sales – cost of goods sold

Gross profit = 2520000 – 1920000

= 600000

Gross profit ratio = 600000/2520000*100

= 23.81%

2. Net Profit Ratio = (NP / Net Sales)* 100

Net Profit Ratio = 360000/2520000*100

14.29%

3) Return on Total Assets = NP/ Total Assets *100

Total assets = Fixed Assets + Current Assets +inventory

14 40000 + 760000 + 800000

= 3000000

Return on Total Assets = 360000/3000000 *100

= 12%

4) Net worth to Debt = Net worth/ Debt

= 1500000/900000* 100

= 1.66 times

5) Working Capital Turnover = Turnover/Working capital

Working Capital = Current Assets – Current Liabilities

= 8,00,000 + 7,60,000 – 6,00,000

Working Capital = 9,60,000

Working Capital Turnover Ratio = **2520000 /960000**

= 2.63 Times.

6) Inventory Turnover Ratio = Turnover / Total Assets* 100

= 1920000/800000= 2.4 times

12) A company's inventory is Rs. 200,000. Total liquid Assets are Rs. 800,000 and quick ratio is 2:1. Calculate current ratio.

Step 1: Calculate Current Liabilities

The quick ratio is defined as:

Quick Ratio = Liquid Assets / Current Liabilities

Given that the quick ratio is 2:1, we can express it as:

$$2 = 800,000 / \text{Current Liabilities}$$

$$\text{Current liabilities} = 800,000 / 2$$

$$\text{Current liabilities} = 400,000$$

Step 2: Calculate Current Assets

Current assets include liquid assets and inventory. Therefore:

Current Assets = Liquid Assets + Inventory

$$\text{Current assets} = 800,000 + 200,000$$

$$\text{Current Assets} = 1,000,000$$

Step 3: Calculate Current Ratio

The current ratio is defined as:

Current ratio = current assets / current liabilities

$$\text{Current ratio} = 1,000,000 / 400,000$$

$$\text{Current ratio} = 2.5:1$$

13) Quick ratio 1.5 current assets is 100,000. Calculate current liabilities is Rs. 40,000. Calculate the value of inventory.

Step 1: Calculate Liquid Assets

The quick ratio is defined as:

Quick ratio = quick assets / current liabilities

$$1.5 = 40,000 / \text{Liquid Assets}$$

$$\text{Liquid Assets} = 1.5 \times 40,000 = 60,000$$

Step 2: Calculate Inventory

Current assets consist of liquid assets and inventory:

Current Assets=Liquid Assets+Inventory

100000 =60000 + inventory

Inventory= 100000-60000

Inventory =40000

14) Current ratio 2.4, current assets rs.1,81,10,400, inventories rs. 79,23,300. Calculate the liquid ratio.

Current ratio = current assets/current liabilities

2.4 = 1,81,10,400/current liabilities

Current liabilities = 1,81,10,400/2.4

Current liabilities = 75,46,000

Liquid ratio = liquid assets/current liabilities

Liquid assets = current assets-inventories

= 1,81,10,400-79,23,300

Liquid assets= 1,01,87,100

Liquid ratio = liquid assets/current liabilities

1,01,87,100/75,46,000

Liquid ratio = 1.35:1

15) A firm has current ratio of 4.5:1 and quick ratio of 3:1. If its inventory is rs.72,000, find out its total current assets and total current liabilities.

Inventory = 4.5 – 3 = 1.5

If inventory is 1.5, current assets = 4.5

If inventory is 1, current assets = 4.5/1.5

If inventory is 72000, current assets = 4.5/1.5*72000

Current assets = 2,16,000

Current liabilities = 2,16,000/4.5

Current liabilities = 48000

16) A business has a current ratio of 4:1 and a quick ratio of 1.2:1. If the working capital is rs. 180000. Calculate the total current assets and inventory.

Working capital is defined as:

Working Capital=Current Assets–Current Liabilities

Working capital = 4 – 1 =3

If working capital is 3 than current assets = 4

If working capital is 1 than current assets =4/3

If working capital is 180000 than current assets =4/3*180000

Current assets = 240000

The current ratio is defined as:

i) Current Ratio= Current Assets /Current Liabilities

4 = 240000 / current liabilities

Current liabilities = 240000/4

Current liabilities= 60000

ii) Quick ratio = quick assets/current liabilities

1.2 = quick assets/60000

Quick assets = 60000*1.2

Quick assets = 72000

iii) Calculate inventory = current assets – liquid assets

Inventory = 240000 – 72000

Inventory = 168000

17) Current assets 85000, inventory 22000, prepaid expenses Rs.2000 working capital 45000.calculate quick ratio.

Working capital =current assets –current liabilities

45000= 85000-current liabilities

Current liabilities= 85000-45000

Current liabilities =40000

Quick ratio= quick assets /current liabilities

Quick assets= current assets –inventory-prepaid Expenses

Quick Assets= 85000-22000-2000

Quick Assets=61000

Quick ratio=61000/40000

Quick ratio=1.53:1

18) Working capital 480000, total debts 1600000, long term debt 1000000; inventory 340000; prepaid insurance 20000.calculate liquid ratio.

Current liabilities= total debts –long term debts

C.L= 1600000-1000000

C.L= 600000

Current assets =working capital +current liabilities

CURRENT assets=480000+600000

Current Assets=1080000

Calculate liquid ratio= liquid Assets/current liabilities

Liquid Assets=Current assets-inventory-prepaid insurance

Liquid assets= 1080000-340000-20000

Liquid assets=720000

Liquid ratio=720000/600000

Liquid ratio=1.2

19) A firm had current assets of rs.4, 10,000. It then paid trade payables of rs.50, 000. After this payment, the current ratio was 2.4:1. Ascertain the amount of current liabilities and working capital after the payment.

Current ratio = current assets/current liabilities

Current assets = 4, 10,000

Current assets after the payment of 4, 10,000 – 50,000 =360000

Current liabilities = 360000/2.4

Current liabilities = 1, 50,000

Working capital = current assets – current liabilities

Working capital = 3, 60,000 – 1, 50,000

Working capital = 2, 10,000

20) Honda tld. Has inventory of rs. 20,000. Total liquid assets are rs.1, 00,000 and quick ratio is 2:1. Calculate current ratio.

Calculate Current liabilities =

Quick ratio = liquid assets/ current liabilities

$$2 = 100000/\text{current liabilities}$$

$$\text{Current liabilities} = 100000/2$$

$$\text{Current liabilities} = 50000$$

Calculate current assets = liquid assets + inventory

$$\text{Current assets} = 100000 + 20000$$

$$\text{Current assets} = 120000$$

Calculate current ratio = current assets/current liabilities

$$\text{Current ratio} = 120000/50000$$

$$\text{Current ratio} = 2.4:1$$

21) Current liabilities of a company are rs.75000. If current ratio is 4:1 and liquid ratio is 1:1. Calculate value of current assets, liquid assets and inventory.

Current ratio = current assets/current liabilities

$$4 = \text{current assets}/75000$$

$$\text{Current assets} = 75000 * 4$$

$$\text{Current assets} = 300000$$

Liquid ratio = liquid assets/current liabilities

$$1 = \text{liquid assets}/75000$$

$$\text{Liquid assets} = 75000 * 1$$

$$\text{Liquid assets} = 75000$$

Inventory = current assets – liquid assets

$$\text{Inventory} = 300000 - 75000$$

$$\text{Inventory} = 225000$$

22)

Particular	Assets	Liabilities
Cash	2400	
Account receivables	7800	
Inventory	11200	
Property, plant equipment	34000	
Investment	5100	
Other assets	1700	
Account payables		6800
Short term debts		4100
Long term debts		26200
Accrued expenses		1700
Other liabilities		920
Equity		22480
Total	62200	62200

Calculate: -

- 1) Calculate the current ratio.
- 2) Determine the debt-to-equity ratio.
- 3) What is quick ratio.
- 4) Compute the gross profit margin has a revenue of 50000 and COGS of 30000.
- 5) Find the return on assets (ROA) for balance sheet assuming a net income is 10000.

1) Current Ratio

Formula:

Current Ratio=Current Assets/Current Liabilities

Current Assets: Cash (2400) + Receivables (7800) + Inventory (11200)
=21,400

Current Liabilities: Accounts Payables (6800) + Short-term Debts (4100) +
Accrued Expenses (1700) + Other Liabilities (920) = **13,520**

Current Ratio=21,400/13,520

≈1.58:1

2) Debt-to-Equity Ratio

Formula:

$$\text{Debt-to-Equity Ratio} = \text{Total Debt} / \text{Equity}$$

$$\text{Total Debt (Liabilities)} = 39,720$$

$$\text{Equity} = 22,480$$

$$\text{D/E Ratio} = 39,720 / 22,480 \approx 1.77:1$$

3) Quick Ratio

Formula:

$$\text{Quick Ratio} = \text{Current Assets} - \text{Inventory} / \text{Current Liabilities}$$

$$\text{Quick Assets} = \text{Cash (2400)} + \text{Receivables (7800)} = 10,200$$

$$\text{Current Liabilities} = 13,520$$

$$\text{Quick Ratio} = 10,200 / 13,520$$

$$\approx 0.75:1$$

4) Gross Profit Margin

Formula:

$$\text{Gross Profit Margin} = \text{Revenue} - \text{COGS} / \text{Revenue} \times 100$$

$$\text{Revenue} = 50,000$$

$$\text{COGS} = 30,000$$

$$\text{Gross Profit Margin} = 50,000 - 30,000 / 50,000 \times 100$$

$$= 20,000 / 50,000 \times 100$$

$$= 40\%$$

5) Return on Assets (ROA)

Formula:

$$\text{ROA} = \text{Net Income} / \text{Total Assets} \times 100$$

$$\text{Net Income} = 10,000$$

$$\text{Total Assets} = 62,200$$

$$\text{ROA} = 10,000 / 62,200 \times 100$$

$$\approx 16.07\%$$